

2.8 Subspaces of $\mathbb{R}^n$

I

This section focuses on important sets of vectors in $\mathbb{R}^n$ called subspaces. Often subspaces arise in connection with some matrix A , and they provide useful information about the equation $A\vec{x} = \vec{b}$. The concepts and terminology in this section will be used repeatedly throughout the rest of the book.

Definition

A subspace of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

- The zero vector is in H .
- For each $\vec{u}$ and $\vec{v}$ in H , the sum $\vec{u} + \vec{v}$ is in H .
- For each $\vec{u}$ in H and each scalar c , the vector $c\vec{u}$ is in H .

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In other words; a subspace is closed under both addition and scalar multiplication. It also contains the zero vector.

Example 1

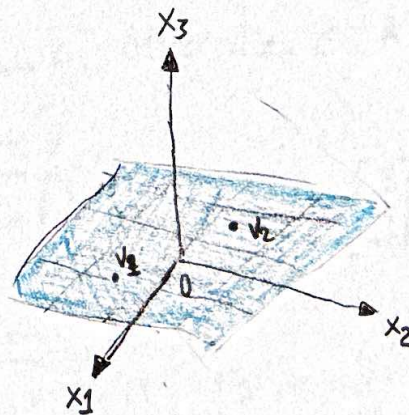
If $\vec{v}_1$ and $\vec{v}_2$ are in $\mathbb{R}^n$ and $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$, then H is a subspace of $\mathbb{R}^n$. To verify the statement, note that the zero vector is in H (because $0\vec{v}_1 + 0\vec{v}_2$ is a linear combination of $\vec{v}_1$ and $\vec{v}_2$). Now take two arbitrary vectors in H , say

$$\vec{u} = s_1\vec{v}_1 + s_2\vec{v}_2 \quad \text{and} \quad t_1\vec{v}_1 + t_2\vec{v}_2 = \vec{v}$$

Then

$$\vec{u} + \vec{v} = (s_1 + t_1)\vec{v}_1 + (s_2 + t_2)\vec{v}_2$$

which shows that $\vec{u} + \vec{v}$ is a linear combination of $\vec{v}_1$ and $\vec{v}_2$ and hence is in H . Also, for any scalar c , the vector $c\vec{u}$ is in H , because $c\vec{u} = c(s_1\vec{v}_1 + s_2\vec{v}_2) = (cs_1)\vec{v}_1 + (cs_2)\vec{v}_2$.



2.8 Subspaces of $\mathbb{R}^n$ (cont'd)

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Example 2

A line L not through the origin is not a subspace, because it doesn't contain the origin which is required per the definition.

Example 3

For $\vec{v}_1, \dots, \vec{v}_p$ in $\mathbb{R}^n$, the set of all linear combinations of $\vec{v}_1, \dots, \vec{v}_p$ is a subspace of $\mathbb{R}^n$. The verification of this statement is similar to the argument given in Example 1. We shall now refer to $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$ as the subspace spanned (or generated) by $\vec{v}_1, \dots, \vec{v}_p$.

Note that $\mathbb{R}^n$ is a subspace of itself because it meets the three requirements.

Column Space and Null Space of a Matrix

Subspaces of $\mathbb{R}^n$ usually occur in applications and theory in one of two ways. In both cases, the subspace can be related to a matrix.

Definition

The column space of a matrix A is the set $\text{Col } A$ of all linear combinations of the columns of A .

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If $A = [\vec{a}_1 \dots \vec{a}_n]$, with the columns in $\mathbb{R}^m$, then $\text{Col } A$ is the same as $\text{Span}\{\vec{a}_1, \dots, \vec{a}_n\}$. Example 4 shows that the column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$. Note that $\text{Col } A$ equals $\mathbb{R}^m$ only when the columns of A span $\mathbb{R}^m$. Otherwise, $\text{Col } A$ is only part of $\mathbb{R}^m$.

Example 4

Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$, and $\vec{b} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$. Determine whether $\vec{b}$ is in the column space of A .

Solution

$\begin{bmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $A\vec{x} = \vec{b}$ has a solution, $\vec{b}$ is in $\text{Col } A$.

2.8 Subspaces of $\mathbb{R}^n$ (cont'd)

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The solution of Example 4 shows that when a system of linear equations is written in the form $A\vec{x} = \vec{b}$, the column space of A is the set of all $\vec{b}$ for which the system has a solution.

Definition

The null space of a matrix A is the set $\text{Nul } A$ of all solutions of the homogeneous equation $A\vec{x} = \vec{0}$.

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When A has n columns, the solution of $A\vec{x} = \vec{0}$ belongs to $\mathbb{R}^n$, and the null space of A is a subset of $\mathbb{R}^n$. In fact, $\text{Nul } A$ has the properties of a subspace of $\mathbb{R}^n$. (p. 179)

Theorem 12

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of a system $A\vec{x} = \vec{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

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Proof

The zero vector is in $\text{Nul } A$ because $A\vec{0} = \vec{0}$. To show the other two properties, take any $\vec{u}$ and $\vec{v}$ in $\text{Nul } A$. Suppose

$$A\vec{u} = \vec{0}, \quad A\vec{v} = \vec{0}$$

then

$$A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v} = \vec{0} + \vec{0} = \vec{0}$$

Also, for any scalar c

$$A(c\vec{u}) = c(A\vec{u}) = c(\vec{0}) = \vec{0}$$

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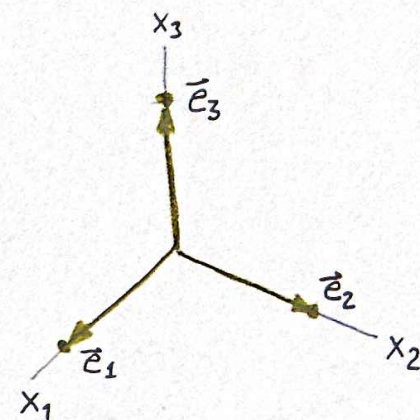
Basis for a subspace

Definition

A basis for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

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$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \vec{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$



The standard basis for $\mathbb{R}^3$

Example 5

The columns of an invertible $n \times n$ matrix form a basis for all of $\mathbb{R}^n$. Because they are linearly independent and span $\mathbb{R}^n$, by the invertible matrix theorem (pp. 145, 189). One such matrix is the $n \times n$ identity matrix. Its columns are denoted by $\vec{e}_1, \dots, \vec{e}_n$ (see figure).

The set $\{\vec{e}_1, \dots, \vec{e}_n\}$ is called the standard basis for $\mathbb{R}^n$.

2.8 Subspaces of $\mathbb{R}^n$ (cont'd)

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Example 6

Find a basis for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution

Row reduction gives

$$A \sim R = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Write up the solution of $A\vec{x} = \vec{0}$ in parametric vector form.

$$\begin{aligned} x_1 - 2x_2 + 0x_3 - 1x_4 + 3x_5 &= 0 \\ 1x_3 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

Identifying x_1 and x_3 as pivot columns.

$$x_1 = 2x_2 + 1x_4 - 3x_5$$

$$x_3 = -2x_4 + 2x_5$$

$$x_2 = r, \quad x_4 = s, \quad x_5 = t$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix}$$

$$x_2 = r, \quad x_4 = s, \quad x_5 = t$$

$$\vec{x} = \begin{bmatrix} 2r + s - 3t \\ r \\ -2s + 2t \\ s \\ t \end{bmatrix} = r \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} = r \vec{u} + s \vec{v} + t \vec{w}$$

The set $\{\vec{u}, \vec{v}, \vec{w}\}$ is a basis for $\text{Nul}(A)$.

2.8 subspaces for $\mathbb{R}^n$ (cont'd)

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Theorem 13

The pivot columns of a matrix A form a basis for the column space of A .

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Warning: Be careful not to use the pivot columns of a reduced matrix as basis for $\text{Col } A$.

The pivot columns you identify are to be taken as reference to the original matrix.

Example 8

Find a basis for $\text{Col } A$ for

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

Solution

$$A \sim R = \begin{bmatrix} \textcircled{1} & 0 & -3 & 5 & 0 \\ 0 & \textcircled{1} & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$

Identifying x_1, x_2 and x_5 as pivot columns.

This gives

$$\vec{u} = \begin{bmatrix} 1 \\ -2 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 3 \\ -2 \\ 3 \\ 4 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} -9 \\ 2 \\ 1 \\ -8 \end{bmatrix}$$

as pivot vectors. Thus

$$\{\vec{u}, \vec{v}, \vec{w}\}$$

forms a basis for $\text{Col } A$.

$$\begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

$$R_2 \leftarrow R_2 \cdot -\frac{1}{2}$$

$$\begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ 1 & 1 & -1 & 4 & -1 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ 0 & -2 & -4 & 2 & 8 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ 0 & -2 & -4 & 2 & 8 \\ 0 & -3 & -6 & 3 & 19 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

$$R_4 \leftarrow R_4 - 3R_1$$

$$\begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ 0 & -2 & -4 & 2 & 8 \\ 0 & -3 & -6 & 3 & 19 \\ 0 & -5 & -10 & 5 & 19 \end{bmatrix}$$

$$R_4 \leftarrow R_4 - (R_2 + R_3)$$

$$\begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ 0 & -2 & -4 & 2 & 8 \\ 0 & -3 & -6 & 3 & 19 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \leftarrow R_2 \cdot (-\frac{1}{2})$$

$$\begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ 0 & 1 & 2 & -1 & -4 \\ 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$

$$\begin{bmatrix} 1 & 0 & -3 & 5 & 3 \\ 0 & 1 & 2 & -1 & -4 \\ 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_3$$

$$R_2 \leftarrow R_2 + 4R_3$$

$$\begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 \ x_2 \ x_3 \ x_4 \ x_5$$

x_1, x_2, x_5 is pivot

x_3, x_4 free

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2.8 Subspaces of $\mathbb{R}^n$ (cont'd)

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Practice problems

1. Let $A = \begin{bmatrix} 1 & -1 & 5 \\ 2 & 0 & 7 \\ -3 & -5 & -3 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix}$ Is $\vec{u}$ in $\text{Col } A$ or $\text{Nul } A$?

$$[A | \vec{u}] = \left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 2 & 0 & 7 & 3 \\ -3 & -5 & -3 & 2 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_3 \leftarrow R_3 + 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 0 & 2 & -3 & 17 \\ 0 & -8 & 12 & -19 \end{array} \right]$$

$$R_3 \leftarrow R_3 + 4R_2$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 0 & 2 & -3 & 17 \\ 0 & 0 & 0 & 7 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot 1/2$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 0 & 1 & -3/2 & 17/2 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & 0 \\ 0 & 1 & -3/2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \leftarrow R_1 + R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 7/2 & 0 \\ 0 & 1 & -3/2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

Row reduction of the augmented matrix reveals an inconsistent system.

$\vec{u}$ is not in $\text{Col } A$.

To see if $\vec{u}$ is in $\text{Nul } A$, we can solve the equation

$$A\vec{u} = \vec{0}$$

$$\begin{bmatrix} 1 & -1 & 5 \\ 2 & 0 & 7 \\ -3 & -5 & -3 \end{bmatrix} \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\vec{u}$ is in $\text{Nul } A$

2. Given $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ Find a vector in $\text{Nul } A$ and in $\text{Col } A$.

A vector in $\text{Col } A$ is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\text{Nul } A$:

$$0x_1 + x_2 + 0x_3 = 0 \Rightarrow x_2 = 0$$

$$0x_1 + 0x_2 + x_3 = 0 \Rightarrow x_3 = 0$$

$x_1 = \text{free variable}$

$$\vec{x} = \begin{bmatrix} s \\ 0 \\ 0 \end{bmatrix} = s \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\vec{u}}$$

$\vec{u}$ is in $\text{Nul } A$.